Radiation modeling

palmgroup

Institute of Meteorology and Climatology, Leibniz Universität Hannover

Radiation modeling

Material:

Mohamed Hefny Salim, PhD
Urban Ecosystem Science
Technische Universität Berlin

Presenter:

Dr. Sebastian Schubert
Urban Ecosystem Science
Technische Universität Berlin

Content

- Introduction
- Radiation models in PALM
  - Simple radiation models
  - Sophisticated radiation models
  - Models for non-building resolving simulations
- Radiative interactions in urban areas
- Representation of radiative transfer model
- Usage and special features
- Example

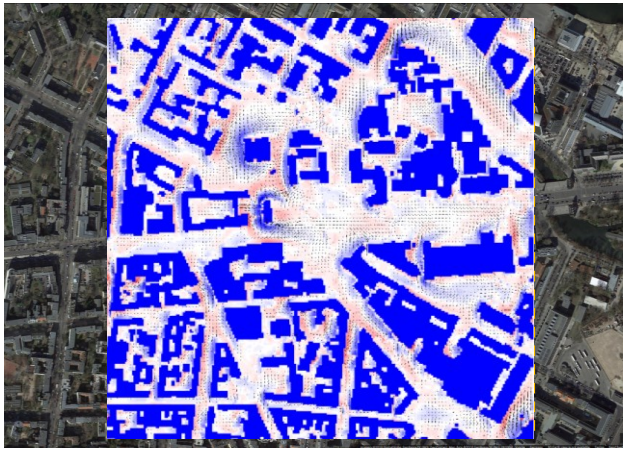
Introduction

Radiation, why should we care?

Inputs:
Terrain data, Buildings, Vegetation, Meteorology, etc.

Outputs:
Wind field (*u*, *v*, *w*), Momentum fluxes, etc.

Applications:
Dynamic effect of obstacles, Wind comfort, Pollutant dispersion, etc.

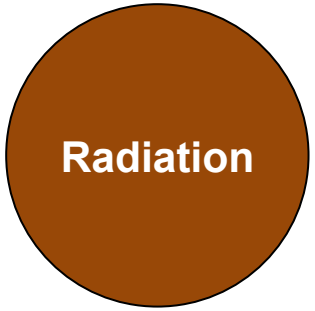


Something is missing?
Air temperature, surface temperature, etc.

Inputs + Solar radiation

When to use a radiation model?

- ✓ Land surface module
- ✓ Urban surface module
- ✓ Chemistry module
- ✓ Plant canopy module
- ✓ Cloud microphysics
- ✓ Multi-agent system
- ✓ Human biometeorology
- ✓ Indoor climate and building energy demand



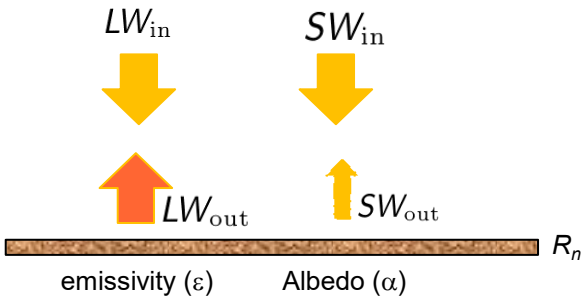
When not then?!

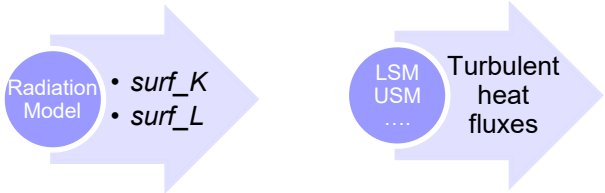
Radiation models in PALM

Simple models

- Constant radiation „model”
  - constant net radiation at the surface: $R_n = const.$
- Clear-sky model
  - Very simple parameterization of fluxes
  - No direct heating of air
  - Broadband albedo only
- External radiation
  - Radiative forcing (short- and longwave downwelling radiation) from a driver

$$R_\mathrm{n} = SW_\mathrm{in} - SW_\mathrm{out} + LW_\mathrm{in} - LW_\mathrm{out}$$





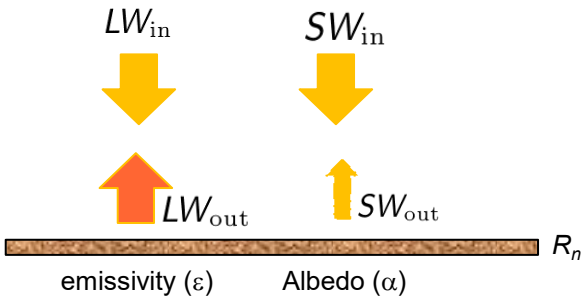
Radiation modeling

Radiation models in PALM

Sophisticated models

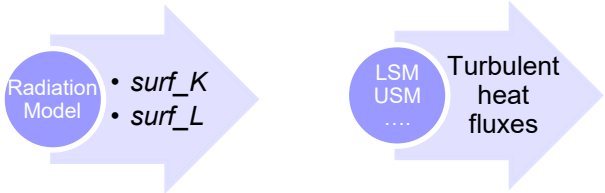
- RRTMG
  - Comprehensive radiation scheme
  - Single-column model for each vertical column of PALM
  - Coupled to PALM (black box)
- TenStream
  - 3D radiation solver
  - Coupled to PALM (black box)

$$R_n = SW_{in} - SW_{out} + LW_{in} - LW_{out}$$



Model for non-building resolving simulations

- DCEP
  - The urban Double Canyon Effect Parametrization scheme (DCEP)



Simple radiation models

Clear-sky model

Radiative budget equation only at the surface:

$$R_\mathrm{n} = \boxed{SW_\mathrm{in}} - \boxed{SW_\mathrm{out}} + \boxed{LW_\mathrm{in}} - \boxed{LW_\mathrm{out}}$$

$$SW_\mathrm{in} = S_0\ \tau\ \sin(\Psi)$$

$S_0 = 1368\ \mathrm{W\ m^{-2}}$: Solar constant $\quad \tau = 0.6 + 0.2\ \sin(\Psi)$ $\quad \Psi$: Zenith angle (°)

$$SW_\mathrm{out} = \alpha\ SW_\mathrm{in}$$

α: surface albedo

$$LW_\mathrm{in} = \epsilon_\mathrm{atm}\ \sigma\ T_1^4$$

$\epsilon_\mathrm{atm} = 0.8$: Emissivity of the atmosphere $\quad T_1$: Temperature at first grid level

$\sigma = 5.67 \cdot 10^{-8}\ \mathrm{W\ m^{-2}\ K^{-4}}$: Stefan-Boltzmann constant

$$LW_\mathrm{out} = \epsilon\ \sigma\ T_0^4$$

$\epsilon = 1$: surface emissivity $\quad T_0$: Skin-temperature

Sophisticated radiation models

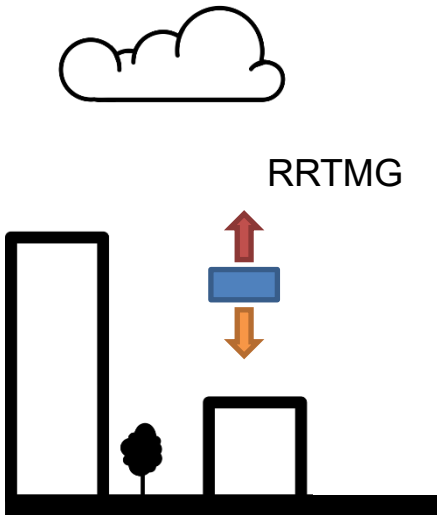
Radiation modeling

Sophisticated radiation models

RRTMG model

Rapid Radiative Transfer Model (for Global Climate Models) http://rtweb.aer.com

- Shipped and coupled to PALM (external library)
- Applied as column model at each vertical column
- Calculates radiative fluxes and heating rates for each grid volume
- Takes into account warm clouds (not ice-clouds)
- Works with bulk microphysics and Lagrangian cloud model



Limitations

- Does not work well with non-stratiform clouds
- Needs considerable computational resources
- Information on the upper atmosphere and trace gases must be provided

Radiation modeling

Sophisticated radiation models

RRTMG usage:

- RRTMG library must be installed, better integration in 25.10
- Re-compile PALM with pre-processor directive `__rrtmg`
- Provide LW/SW input files for RRTMG (*_rlw.nc, *_rwl.nc)

Inputs:

- day of the year (DOY), time (UTC)
- Profiles
  - Hydrostatic pressure
  - Temperature
  - Trace gas volume mixing ratios (H_2O, O_3, CO_2, CH_4, N_2O, O_2, CFC_{11}, etc.)
- Surface values:
  - Surface emissivity
  - Temperature
  - Albedo* ($\alpha_{sw,direct}$,$\alpha_{sw,diffuse}$,$\alpha_{lw,direct}$,$\alpha_{lw,diffuse}$)

* based on Briegleb et al. (1986) and Briegleb (1992)

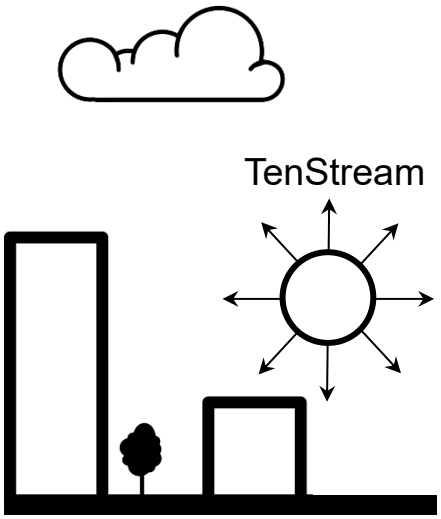
Radiation modeling

Sophisticated radiation models

TenStream radiative transfer model

Parallel approximate solver for the full 3-D radiative transfer equation

- Coupled to PALM (external library)
- 3-D propagation of radiation in the atmosphere.
- 3-D radiative fluxes and heating rates for each grid volume
- Interactions with atmospheric constituents, such as water vapor, fog, and clouds.
- Dynamic heterogeneities (moving clouds or fog).



Limitations

- Needs considerably more computational resources than RRTMG
- Information on the background atmosphere must be provided
- Look-Up-Tables (LUT): voxel radiation-transport coefficients required for performing the radiative transfer processes.

Radiation modeling

Sophisticated radiation models

TenStream usage:

- TenStream library must be installed (see installation instructions)
- Re-compile PALM with pre-processor directive `__tenstream`
- Provide LUT and background atmosphere input files

Inputs:

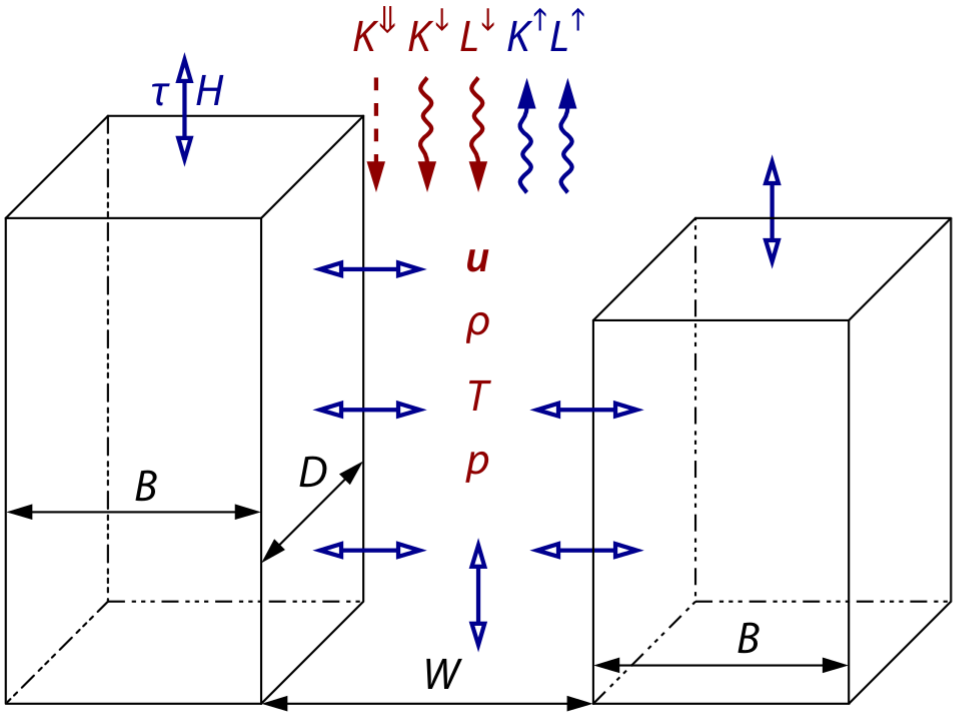
- day of the year (DOY), time (UTC)
- Profiles
  - Hydrostatic pressure
  - Temperature
  - Trace gas volume mixing ratios (H_2O, O_3, CO_2, CH_4, N_2O, O_2, CFC_{11}, etc.)
- Surface values:
  - Surface emissivity
  - Temperature
  - Albedo* ($\alpha_{sw,direct}$,$\alpha_{sw,diffuse}$,$\alpha_{lw,direct}$,$\alpha_{lw,diffuse}$)

* based on Briegleb et al. (1986) and Briegleb (1992)

Radiation for non-building resolving simulations

PALM/DCEP

DCEP: the urban Double Canyon Effect Parametrization scheme



B: building width W: street width D: canyon length

Input:

$L^{\downarrow}$: longwave rad. (down)

$K^{\downarrow,\Downarrow}$: shortwave rad. (down)

$\boldsymbol{u}$: wind velocity

ρ: air density

T: air temperature

p: air pressure

Output:

τ: momentum flux

H: sensible heat flux

$L^{\uparrow}$: longwave rad. (up)

$K^{\uparrow}$: shortwave rad. (up)

Radiation modeling

Radiation for non-building resolving simulations

Urban parametrization scheme DCEP

- multi-layer
- longwave and shortwave radiation with reflections
- momentum and heat fluxes, corresponding TKE production
- requires input parameters
  - urban or vegetation fraction
  - building height distribution
  - street and building width

  alternative: use land use or local climate zone maps with typical values

- currently
  - no water and snow storage, no latent heat flux → in development
  - no building interior → developed, not yet implemented in PALM

Radiative transfer in urban area

Radiation modeling

Radiative transfer in urban area

London skyscraper Walkie Talkie

Royal Town Planning Institute: “.. a daily reminder never to let such a planning disaster ever happen again”

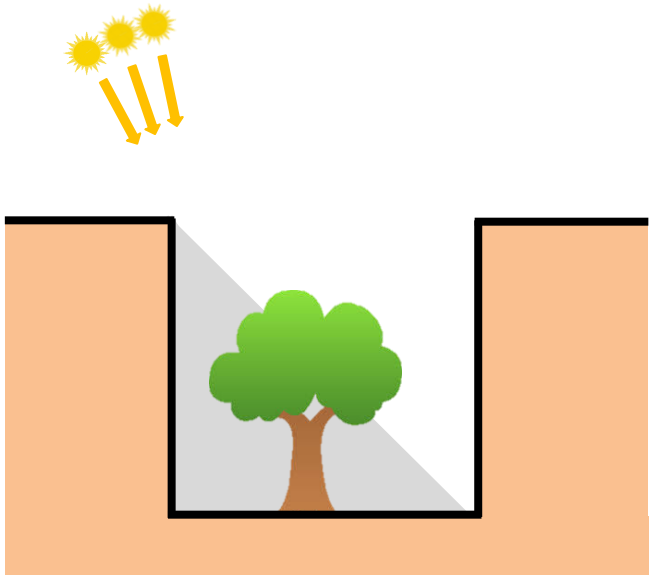
Photos source: http://www.gallopper.com/walkie-talkie-building/

Radiation modeling

Radiative transfer in urban area

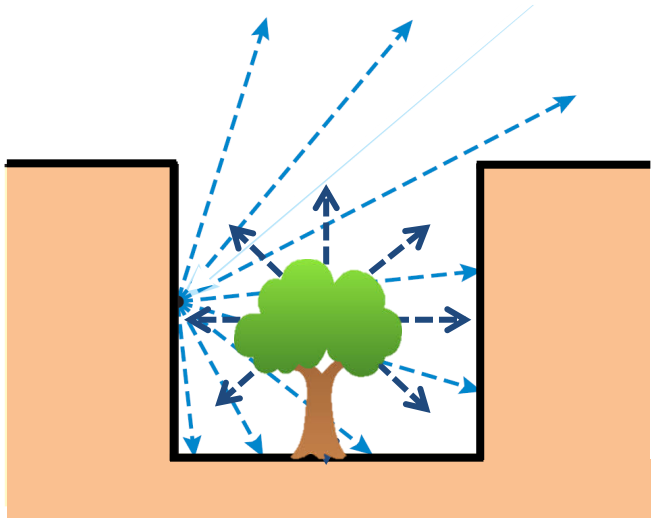
Challenges (1)

- Visibility
  - Sun (shadows)
  - Sky (LW)
- Plant canopy (trees)
  - Transparency (SW)
  - LW absorption



Challenges (2)

- Surface thermal emissions
  - Surface-surface
  - Surface-sky
- Plant canopy (trees) thermal emissions
  - Tree-surface
  - Tree-sky



Challenges (3)

- Reflections
  - SW
  - LW

Something is missing?!

- Plant canopy (trees)
  - SW & LW absorption
  - LW reflections

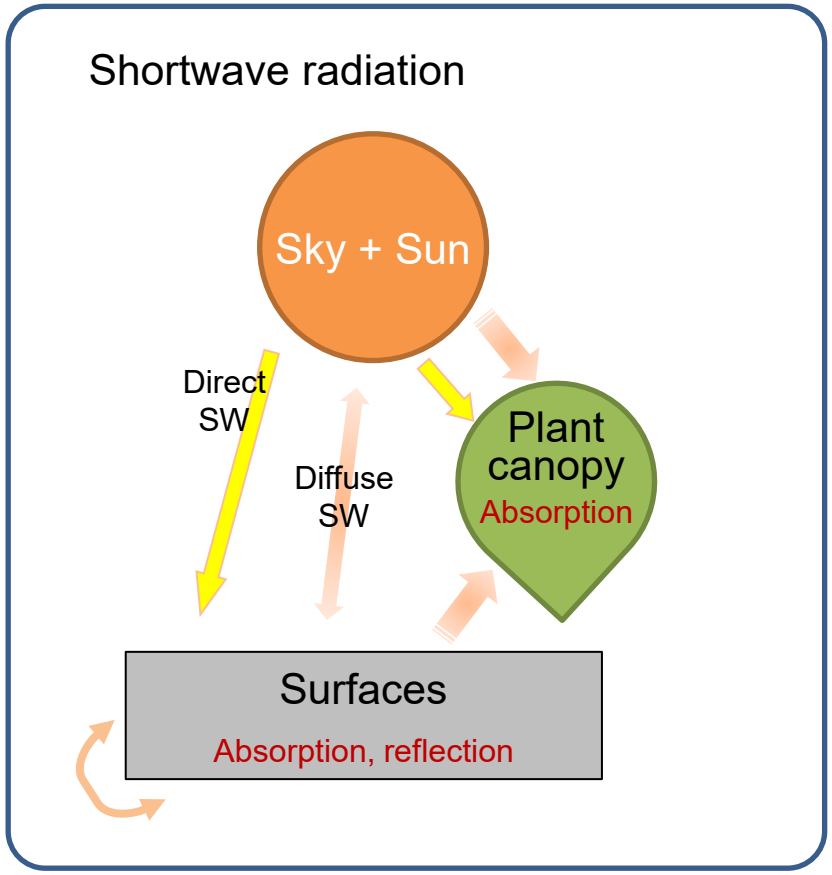
Radiative transfer model (RTM)

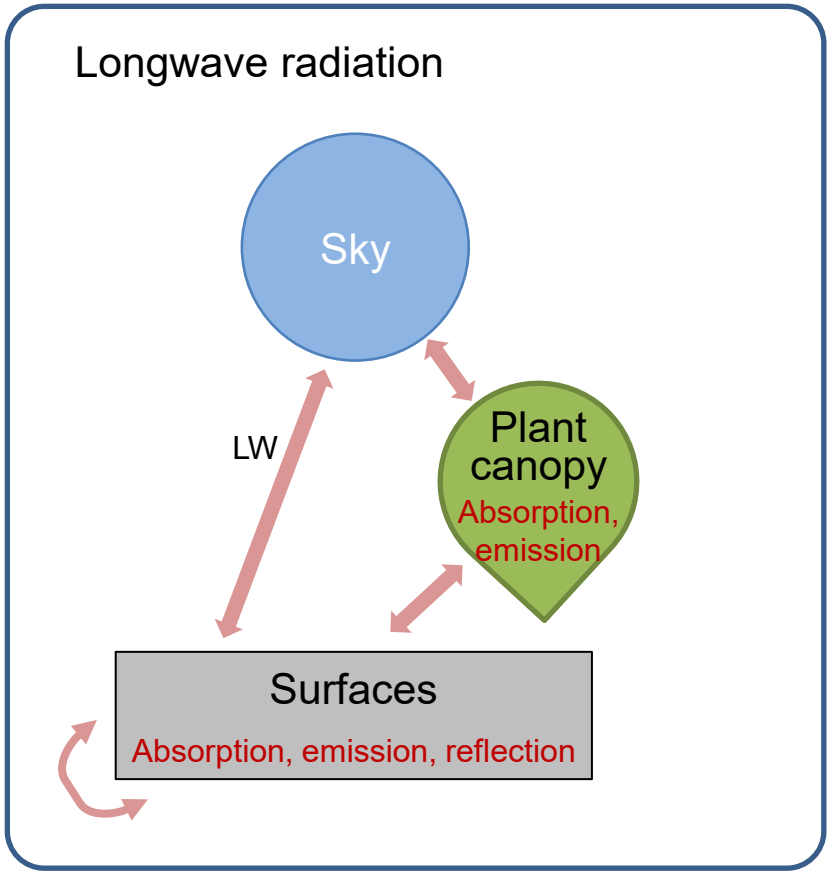
Representation of radiative transfer model

Radiation modeling

Representation of radiative transfer model

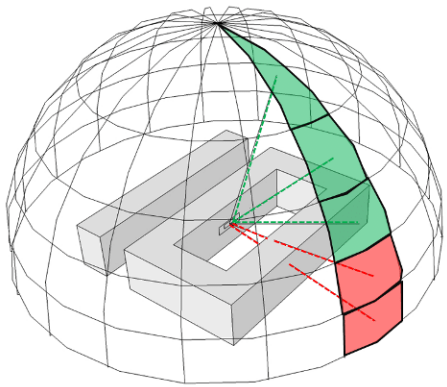
Radiative processes simulated by RTM





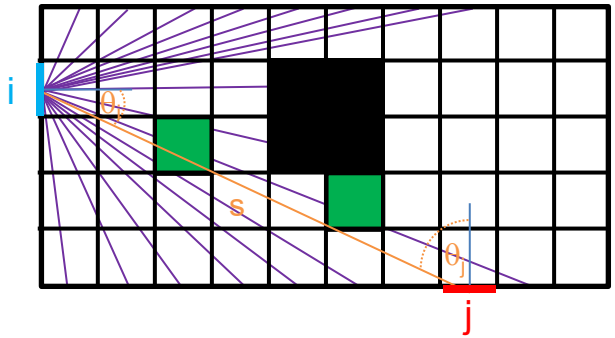
Representing radiative interactions

- View factors
  - Sky view factors (direct SW)
  - Sky view factors (diffuse SW, LW_{sky})
  - Surface view factors (LW: emission, reflections)



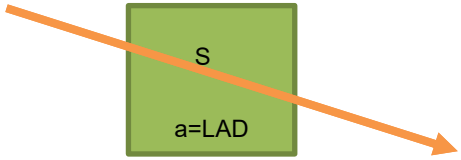
$$F_{i\to j} = \frac{\cos\theta_i \cos\theta_j}{\pi s^2} \qquad \sum_{j=1}^{n} F_{i\to j} = 1$$

- Ray tracing



- Plant canopy sink factors
  - CSF

Transmittance: $T = \frac{\Phi_e^t}{\Phi_e^i} = e^{-\alpha a s}$



Radiation modeling

Representation of radiative transfer model

Problems

- View factors are M~~X~~NY

MANNNN

RTM limitations

- Finite reflection number (4 is fine)
- Diffuse reflection (Lambertian reflectors)
- No interaction with air
- No thermal capacity of plant leaves
- No plant canopy internal interactions
- Surface temperature of leaves is set to air temperature

Radiation modeling

Representation of radiative transfer model

Coupling to RRTMG

RRTMG: 1D radiation above canopy layer

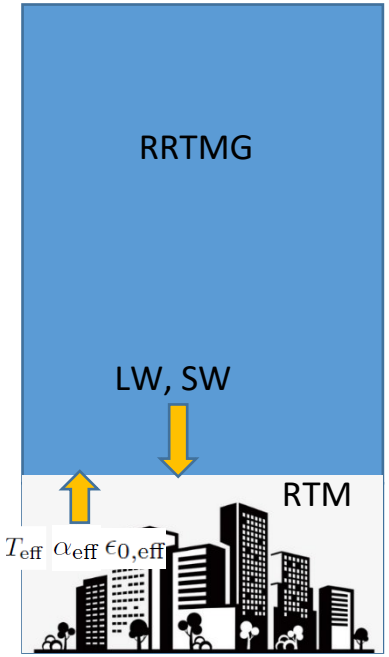
RTM : 3D radiation below canopy layer

Effective radiation surface parameters

$$T_{\text{eff}} \qquad \alpha_{\text{eff}} \qquad \epsilon_{0,\text{eff}}$$

$$SW^{\uparrow} = \alpha_{\text{eff}} SW^{\downarrow}$$

$$LW^{\uparrow} = \epsilon_{\text{eff}} \sigma T_{0,\text{eff}}^4 + (1 - \epsilon_{0,\text{eff}}) LW^{\downarrow}$$



Usage and special features

Radiation modeling

Usage and special features

Radiation namelist

radiation_scheme:
'constant', 'clear-sky', 'external' , 'rrtmg','tenstream'

dt_radiation: timestep of radiation model

albedo_type	Description	broadband	longwave	shortwave	Notes
0	user defined	-	-	-	
1	ocean	0.06	0.06	0.06	
2	mixed farming, tall grassland	0.19	0.28	0.09	
3	tall/medium grassland	0.23	0.33	0.11	
4	evergreen shrubland	0.23	0.33	0.11	
5	short grassland/meadow/shrubland	0.25	0.34	0.14	
6	evergreen needleleaf forest	0.14	0.22	0.06	
7	mixed deciduous forest	0.17	0.27	0.06	
8	deciduous forest	0.19	0.31	0.06	
9	tropical evergreen broadleaved forest	0.14	0.22	0.06	
10	medium/tall grassland/woodland	0.18	0.28	0.06	
11	desert, sandy	0.43	0.51	0.35	
12	desert, rocky	0.32	0.40	0.24	
13	tundra	0.19	0.27	0.10	

Number of reflection steps

```
&radiation_parameters
    radiation_scheme = 'rrtmg',
    dt_radiation = 60.0,
    albedo = 0.2,
    albedo_type = 17,
    constant_albedo = .F.,
    nrefsteps = 4,
    max_raytracing_dist = 200.0,
    min_irrf_value = 0.000001,
    plant_lw_interact, = .T.,
    rad_angular_discretization = .T.,
    radiation_interactions_on = .T.,
    raytrace_discrete_azims = 40,
    raytrace_discrete_elevs = 80,
    raytrace_mpi_rma = .T.,
    skip_time_do_radiation = 0.0,
    surface_reflections = .T.,
    unscheduled_radiation_calls = .T.,
/
```

https://docs.palm-model.org/25.04/Reference/LES_Model/Namelists/#radiation-parameters

Radiation modeling

Usage and special features

Runtime parameters namelist

- outputs
  - Profile data
  - 3D data (volume)
  - Sectional data (e.g. xy)

```
&runtime_parameters

    data_output_pr = !-- 1) profile data
                'u','wu','w"u"',"w*u*','u*2',
                'v','wv','w"v"',"w*v*','v*2',
                'w','w*2','e', 'e*','w*e*',
                'rad_lw_in','rad_lw_out',
                'rad_sw_in','rad_sw_out',
                'wthetav','w"thetav"',"w*thetav*',

    data_output = !-- 2) Volume data
                'theta','theta_av','q','q_av','u','u_av',
                'v','v_av','w','w_av','e','e_av',
                'rtm_rad_pc_inlw','rtm_rad_pc_insw',
                'rtm_rad_pc_inswdir','rtm_rad_pc_inswdif',
                'rtm_rad_pc_inswref',

                    !-- 3) section data (e.g., x,y)
                'rad_net*_xy','rad_net*_xy_av',
                'rad_lw_in*_xy','rad_lw_out*_xy',
                'rad_sw_in*_xy','rad_sw_out*_xy',
                /
```

Surface data output parameters namelist

- Surface outputs
  - NetCDF file
  - vtk files
- Script available to make NetCDF file available in Paraview (palm_srfc2xmf.py in share/utils)

```
&surface_data_output_parameters

   data_output_surf = 'theta_surface', 'theta_surface_av‘,
                      'rad_net', 'rad_net‘_av,
                      'rad_lw_in','rad_lw_in_av',
                      'rad_lw_out','rad_lw_out_av',
                      'rad_lw_dif','rad_lw_dif_av',
                      'rad_lw_ref','rad_lw_ref_av',
                      'rad_lw_res','rad_lw_res_av',
                      'rad_sw_in','rad_sw_out',
                      'rad_sw_dif','rad_sw_ref',
                      'rad_sw_res','rad_sw_dir',

   dt_dosurf = 3600.0,
   averaging_interval_surf = 3600,
   dt_dosurf_av = 3600,
   to_netcdf = .TRUE.,
   to_vtk = .TRUE.,
   skip_time_dosurf = 86400.0,

/
```

https://docs.palm-model.org/25.04/Reference/LES_Model/Namelists/#radiation-parameters

Example

Radiation modeling

Example

Thank you!